He picked up the cover and ran, I chased him till out of the gate. As the driver turned the corner, I noticed three or four men in a jeep. It was a trap.

I have been careful since then. A few casual workers have come home, some with fish and some with fruits; I scared them away.

Muthunayagam told Dhawan bluntly that he would not report to Gowarikar. Dhawan was in a fix. He couldn't let such a senior person go; neither could he reverse the appointment of Gowarikar, who was by no means inferior to Muthunayagam.

Dhawan finally found a way out by creating an Auxiliary Propulsion Systems Unit (APSU), and made Muthunayagam its director. This was the precursor to the Liquid Propulsion Systems Unit (LPSU), which in turn developed into today's Liquid Propulsion Systems Centre (LPSC) at Valiyamala, Thiruvananthapuram. APSU was a notch below the status of an ISRO centre, but the unit director would report directly to the ISRO chairman.

I became part of APSU—and later LPSU—and was working on the Vikas engine. Though it was not yet approved as a project, Dhawan had sanctioned Rs 1 crore from his discretionary fund. While we were putting together the 'one-and-a-half' engine using original material used by SEP, I was working on indigenisation of materials. For I knew we were lucky to have friends like Rene Morin who went out of the way to get me crucial materials and spares from SEP suppliers, this wouldn't be possible every time we wanted to put together a liquid propulsion engine. As for the first piece, I didn't want to experiment with indigenous parts as a failure of a ground test, I knew, would ground Vikas forever.

Rockets have a lot of aluminium alloys. In its purest form, aluminium is not difficult to make, but developing the alloys, with specific ratios that imparted properties to withstand high pressure and temperature, was a tough job. In this case, we needed an alloy called LM-6. While many Indian laboratories at that time were not confident of developing a new material, leave alone for rocketry that was for many still a lofty idea, the Regional Research Laboratory in Pappanamcode, Trivandrum sounded enthusiastic. Given the specifications, the lab did deliver some material, which wasn't bad, but not the best.

A bigger challenge was developing the equivalent of an alloy that the Americans called HS-25 (Hayne's Satellite alloy), and the French called KC-20WN, used to make the combustion chamber nozzle of the liquid propulsion